THE PUSHFORWARD ALONG A PROPER MORPHISM COMMUTES WITH DIRECT SUMS FOR COEFFICIENTS IN Z/NZ

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a quasi-compact base, the fibre dimension bound, the finiteness of the cohomological dimension, and the application of the commutation lemma.

Editorial note. The label lemma-proper-mod-n-direct-sums names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Boundedness of the fibre dimension of a morphism of finite type. Morphisms, the lemma bounding the dimension of the fibres of a morphism of schemes. Supplies the boundedness of the fibre dimension used to get finite cohomological dimension.
- Finiteness of the cohomological dimension of a proper morphism. Same chapter, the lemma bounding the cohomological dimension of a proper morphism in terms of the fibre dimension. Supplies cd(f) $<$ infinity.
- Commutation of the pushforward with direct sums under finite cohomological dimension. Same chapter, the lemma stating that the pushforward with Z/nZ coefficients commutes with direct sums when the cohomological dimension is finite. Supplies the conclusion.

Source reference. Original reference 095U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18786–18786. Prerequisite pointer invoked at proof line(s) 20158; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0F11: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 20074–20074. Prerequisite pointer invoked at proof line(s) 20160; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0A3V: sources/stacks-chow/tags.txt, source lines 12753–12753. Prerequisite pointer invoked at proof line(s) 20155; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes. Let $n \geq 1$ be an integer. Then the functor*

$$Rf_* : D(X_{étale}, \mathbf{Z}/n\mathbf{Z}) \longrightarrow D(Y_{étale}, \mathbf{Z}/n\mathbf{Z})$$

commutes with direct sums.

Proof. It is enough to prove this when Y is quasi-compact. By Morphisms, Lemma 0A3V we see that the dimension of the fibres of $f : X \to Y$ is bounded. Thus Lemma 095U implies that $\mathrm{cd}(f) < \infty$. Hence the result by Lemma 0F11. □